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Thermodynamic modeling of Fe–Ce–Bi system and explanation of interfacial reaction in Fe-45 at%–Ce/Bi couple

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ABSTRACT

Knowledge of phase equilibria of Fe–Ce–Bi is important for understanding respective effects of rare earth, low melting point elements, e.g. Bi, and the interaction between rare earth and low melting point elements in cast iron. By using a CALPHAD approach, the Ce–Bi binary system was assessed on the basis of reported thermodynamic and phase-equilibrium data and then extrapolated to Fe–Ce–Bi ternary system combined with the Fe–Ce and Fe–Bi binary systems assessed in literatures. Meanwhile, the interfacial reaction between Bi and Fe-45 at%–Ce alloy at 773 K was investigated to verify the reliability of thermodynamic calculations. It was clearly observed that a reaction layer containing $CeBi_2$, Fe and liquid (Bi) exists in the couple, indicating the reaction path liquid (Bi)/$CeBi_2$+Fe/Fe-45 at%–Ce in this couple. The reaction path was further explained using a recently reported model by adopting the thermodynamics assessed herein.

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1. Introduction

Elements with a low melting point, including As, Bi, Sn and Sb, often exist as impurity elements in iron and steel and are generally detrimental to the properties of Fe-based alloys. Meanwhile, due to their activated chemical properties, rare earth (RE) elements are often added as the main components of modifiers to the melts of cast iron or steel to improve their mechanical properties [1]. Besides the strengthening effect of refining grain, the addition of RE to an iron melt may also play a deoxidizing or desulfurizing role [2]. Recent research reported that the toughness and hardness of cast iron could benefit from the addition of RE, Al and low melting-point elements (LME) [3]. Therefore, beyond these effects of RE, it is not clear whether the improvement of mechanical properties by the interaction between RE and LME is exclusive to cast iron. In order to better understand the metallurgical role of RE and LME, knowledge of phase equilibria and the thermodynamics of Fe–RE–LME systems are fundamentally necessary.

This paper aims to assess the thermodynamics of the Fe–Ce–Bi ternary system. Of the three boundary binary systems, the Ce–Fe and Bi–Fe binary systems have been well assessed thermodynamically by Su and Tedenac [4] and Boa et al. [5], respectively. Good agreement was carried out between the calculated phase diagrams and experimental data. Figs. 1 and 2 show the calculated phase diagrams of Ce–Fe and Bi–Fe, respectively. No thermodynamic assessment of Ce–Bi binary system has so far been published. The present study focuses first on optimization of the Ce–Bi binary system and then on extrapolation of the Fe–Ce–Bi ternary system based on the established Bi–Fe and Ce–Fe systems. Finally, interfacial reaction in the Fe-45 at%–Ce/Bi couple will be measured and explained with our calculations.

2. Evaluation of experimental information

2.1. Ce–Bi binary systems

The Ce–Bi phase diagram assembled by Gschneidner et al. [6] was based on the boundary data reported by Vogel et al. [7], Pleasance et al. [8] and Yoshihara et al. [9]. Vogel et al. [7] firstly measured the phase equilibrium of Ce–Bi binary system. Ce_3Bi, Ce_4Bi_3, CeBi and $CeBi_2$ were shown to be stable compounds, with Ce_4Bi_3 the only compound in the Ce–Bi binary system that melted congruently. However, his raw materials had been proved to be impure, raising doubt regarding many details of the reported phase diagram. Pleasance et al. [8] studied the solubility of Ce in liquid Bi. Yoshihara et al. [9] determined the complete range of Ce–Bi alloys by means of X-ray diffraction and metallographic techniques. They confirmed that the first compound shown on the Ce-rich side is Ce_2Bi and not Ce_3Bi as Vogel et al. [7] reported, and identified the crystal structure of Ce_5Bi_3 (Mn_5Si_3-type).

In the Ce–Bi binary system, five compounds Ce_2Bi, Ce_5Bi_3, Ce_4Bi_3, CeBi and $CeBi_2$ were reported [7–9] and all were formed through peritectic reactions except Ce_4Bi_3. The homogeneity ranges of these compounds in the Ce–Bi system are less than

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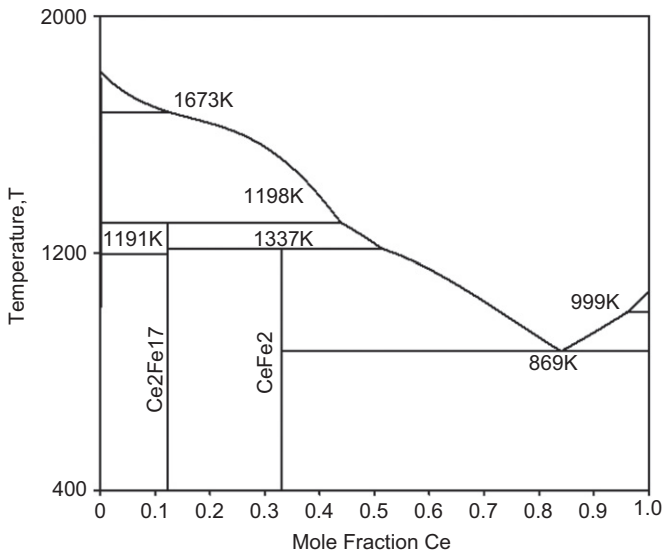

Fig. 1. The Ce–Fe phase diagram calculated from Su and Tedenac. [4].

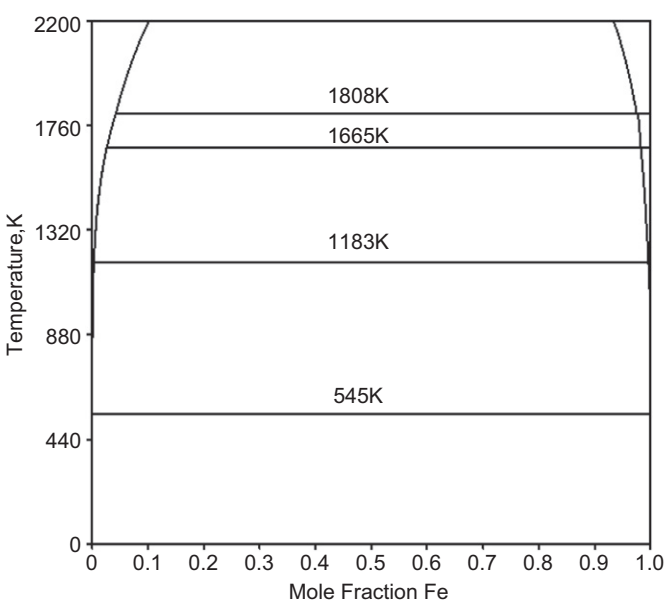

Fig. 2. The Bi–Fe phase diagram calculated from Boa et al. [5].

± 0.5 at%, so they will be treated as stoichiometric compounds. Besides, there are other phases—liquid, γCe (FCC_A1), δCe (BCC_A2) and Bi (Rhomb_A7)—present in this system.

The heats of formation of intermetallic compounds in the Ce–Bi system were measured at 300 K by Borzone et al. [10] using an isoperibol direct calorimeter. The following values were measured for the various compounds: Ce_2Bi, −81.6 kJ/mol; Ce_5Bi_3, −91.7 kJ/mol; Ce_4Bi_3, −101.3 kJ/mol; CeBi, −114.7 kJ/mol and $CeBi_2$, −77.5 kJ/mol. Dong et al. [11] estimated the mixing enthalpies of liquid Ce–Bi alloys by using Miedema's model. It was indicated that the mixing enthalpy of the liquids took a negative value across the whole composition range.

2.2. Fe–Ce–Bi ternary system

Fe–Bi and Fe–Ce systems have been assessed thermodynamically. There are no intermetallic compound phases in the binary constituent Fe–Bi. The Fe–Ce system has two stoichiometric phases: $Fe_{17}Ce_2$ and Fe_2Ce. For details of these evaluations, Su and Tedenac [4] and Boa et al. [5], respectively, can be referred. These recent assessments of the Fe–Ce [4] and Fe–Bi [5] systems will be adopted in our extrapolation.

No experimental data on phase equilibria or thermodynamics are available for the ternary Fe–Ce–Bi system. However, according to the available information, the ternary Fe–Ce–Bi system has the following 13 phases: liquid, γ-Ce (FCC_A1), δ-Ce (BCC_A2), γ-Fe (FCC_A1), α-Fe (BCC_A2), Bi (Rhomb_A7), Ce_2Bi, Ce_5Bi_3, Ce_4Bi_3, CeBi, $CeBi_2$, $Fe_{17}Ce_2$, and Fe_2Ce. In extrapolation, it is assumed that no ternary phase is stable in the Fe–Ce–Bi system, and that the solubility of ternary elements in binary compounds is negligible. Further information about the crystalline structures of the phases involved is presented in Table 1.

Table 1
Crystal structure data of the intermetallic compounds in the Fe–Ce–Bi ternary system.

Phase	Prototype	Space group	References
Ce_2Bi	La_2Sb	$tI12$, $(I4/mmm)$	[9]
Ce_4Bi_3	Th_3P_4	$cI28$, $(I-43d)$	[9]
CeBi	NaCl	$cF8$, $(Fm-3m)$	[9]
Ce_5Bi_3	Mn_5Si_3	$hP16$, $(P6_3/mcm)$	[9]
$CeBi_2$	*	aP^*, *	[9]
Ce_2Fe_{17} rhom	$Zn_{17}Th_2$,	$hr57$, $(R-3m)$	[20]
Ce_2Fe_{17} hex	Th_2Ni_{17},	$hP38$, $(P6_3/mmc)$	[21]
$CeFe_2$	$MgCu_2$,	$cF24$, $(Fd-3m)$	[22]

3. Thermodynamic models

3.1. Pure elements

The standard element reference (SER) state, i.e. the stable structure of the element at 298.15 K and 100 kPa, is used as the reference states. The Gibbs energy of each pure element, i, in the φ status is given as

$$^0G_i^{\varphi}(T) = G_i^{\varphi} - H_i^{SER} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1)$$

where H_i^{SER} is the enthalpy of element i in its SER state and T is the temperature. The Gibbs energy of the element i, $^0G_i^{\varphi}(T)$, with structure φ referred to the enthalpy of its stable state at 298.15 K and 100 kPa, is denoted by $GHSER_i$, i.e.,

$$GHSER_i = {}^0G_i^{\varphi}(T) = G_i^{\varphi} - H_i^{SER} \quad (2)$$

In present work, $^0G_{Bi}^{\varphi}(T)$, $^0G_{Ce}^{\varphi}(T)$ and $^0G_{Fe}^{\varphi}(T)$ are taken from the SGTE (Scientific Group Thermodata Europe) database [12].

3.2. Substitution solutions

The liquid, FCC_A1, BCC_A2, and Rhomb_A7 phases are modeled by the substitutional solution model (Bi, Ce, Fe), of which the molal Gibbs energy is expressed as follows:

$$G_m^{\varphi} = \sum_{i=\mathrm{Bi,Ce,Fe}} x_i {}^0G_i^{\varphi} + RT \sum_{i=\mathrm{Bi,Ce,Fe}} x_i \ln x_i + {}^EG^{\varphi} \quad (3)$$

where $^0G_i^{\varphi}$ represents the molar Gibbs energy of pure substances i in the φ state, and $^EG^{\varphi}$ is excess Gibbs energy, expressed by the Redlich–Kister polynomial [13]:

$$^EG^{\varphi} = x_{Bi}x_{Ce} \sum_{i=0,1\ldots} {}^iL_{Bi,Ce}^{\varphi}(x_{Bi}-x_{Ce})^i + x_{Bi}x_{Fe} \sum_{i=0,1\ldots} {}^iL_{Bi,Fe}^{\varphi}(x_{Bi}-x_{Fe})^i$$

$$+x_{Bi}x_{Ce}\sum_{i=0,1\ldots}{}^{i}L^{\varphi}_{Ce,Fe}(x_{Ce}-x_{Fe})^{i}+x_{Bi}x_{Ce}x_{Fe}L^{\varphi}_{Bi,Ce,Fe} \quad (4)$$

where ${}^{i}L^{\varphi}_{Bi,Ce}$, ${}^{i}L^{\varphi}_{Bi,Fe}$ and ${}^{i}L^{\varphi}_{Ce,Fe}$ are the binary interaction parameters, of which ${}^{i}L^{\varphi}_{Bi,Fe}$ and ${}^{i}L^{\varphi}_{Ce,Fe}$ are taken from the Ce–Fe [4] and Bi–Fe [5] binary systems, respectively, and ${}^{i}L^{\varphi}_{Bi,Ce}$ is to be assessed in the present study. The ternary interaction parameter $L^{\varphi}_{Bi,Ce,Fe}$ is set as zero, owing to a lack of sufficient information including phase equilibria and thermodynamics about the Fe–Ce–Bi ternary system. The binary interaction parameters ${}^{i}L_{\varphi}$ can be temperature dependent as follows:

$${}^{i}L^{\varphi}=A_i+B_iT+C_iT\ln T \quad (5)$$

where A_i, B_i and C_i are interaction parameters for the Ce–Bi binary system to be optimized in the present study.

The Rhomb_A7 structure for pure Ce is not stable. According to estimation of Guilermet [14] the following lattice stability was given:

$$G(\text{Rhomb_}A7,Ce;0)=GHSER_{Ce}+5000.0 \quad (6)$$

In the γ-Ce (FCC_A1) and Bi (Rhomb_A7) phases, no remarkable solubility was measured. So a large positive interaction parameter is assigned to these two phases, i.e.

$$L(FCC_A1,\text{Bi},\text{Ce}:VA;0)=100{,}000 \quad (7)$$

$$L(\text{Rhomb_A7},\text{Bi},\text{Ce};0)=100{,}000 \quad (8)$$

In addition, the FCC_A1 (γ-Fe) and BCC_A2 (α-Fe) phases exhibit antiferromagnetic and ferromagnetic behaviors, respectively. According to Hillert and Jarl [15], a magnetic contribution to the corresponding Gibbs energy should be involved, which is described by the following relation:

$${}^{mag}G^{\varphi}_{m}=RT\ln(\beta^{\varphi}+1)f(\tau) \quad (9)$$

$$\tau=T/T^{\varphi}_{c} \quad (10)$$

where T^{φ}_{c} is the critical Curie or Neel temperature and β^{φ} is the average Bohr magneton number per mole of atoms. The term $f(\tau)$ represents a polynomial and can be found in the literature [15]. In order to describe the composition dependence of the magnetic ordering temperature, the magnetic moment and the ferromagnetic transition temperature were assumed to decrease linearly from pure iron.

3.3. Stoichiometric intermetallic compounds

Ce_2Bi, Ce_5Bi_3, Ce_4Bi_3, CeBi and $CeBi_2$ are treated as stoichiometric compounds because of their narrow homogeneity. Their molar Gibbs energies are calculated by

$$G^{Ce_ABi_B}_{m}=\frac{A}{A+B}{}^{0}G_{Ce}+\frac{B}{A+B}{}^{0}G_{Bi}+a_1+b_1T \quad (11)$$

where a_1 and b_1 are parameters to be optimized.

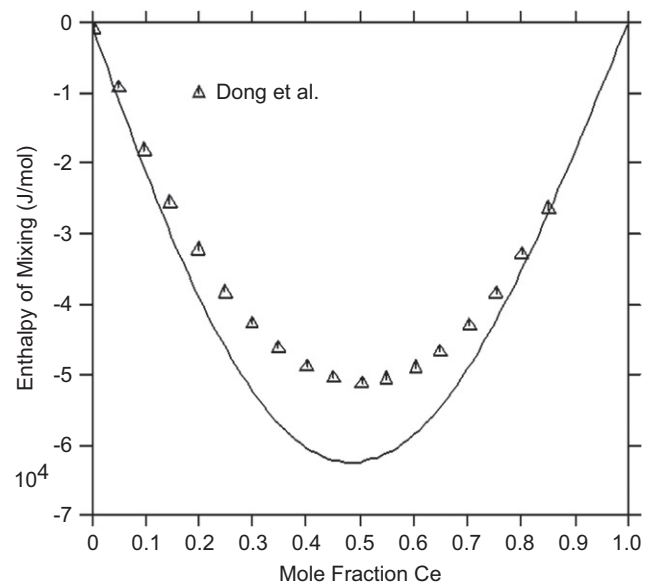

Fig. 3. Calculated enthalpy of mixing of liquid in the Ce–Bi system at 298 K in comparison with the prediction of Miedema's model [11].

Table 2
The thermodynamic interaction parameters of ternary Fe–Ce–Bi system.

Phase	Interaction parameter		References
Liquid	L (liquid,Bi,Ce;0)	$-249267.26+29.87*T+0.198*T*\ln(T)$	This work
	L (liquid,Bi,Ce;1)	$-18760.41+18.97*T$	This work
	L (liquid,Bi,Ce;2)	47458.51	This work
	L (liquid,Ce,Fe;0)	$5881.8-8.5586*T$	[4]
	L (liquid, Ce, Fe; 1)	$12{,}585-20.1552*T$	[4]
	L (liquid,Ce,Fe;2)	2591.0	[4]
	L (liquid,Bi,Fe;0)	$62327.74-4.3831*T$	[5]
	L (liquid,Bi,Fe;1)	-3362.48	[5]
FCC_A1	L (FCC_A1,Bi,Ce:VA;0)	100,000	This work
	L (FCC_A1,Ce,Fe:VA;0)	59316.5	[4]
	L (FCC_A1,Bi,Fe:VA;0)	44,480	[5]
BCC_A2	L (BCC_A2,Bi,Ce:VA;0)	$-182333.46+5.65*T$	This work
	L (BCC_A2,Ce,Fe:VA;0)	$3736.6+35.2022*T$	[4]
	L (BCC_A2,Bi,Fe:VA;0)	$39295.21+3.388*T$	[5]
Rhomb_A7	L (Rhomb_A7,Bi,Ce:VA;0)	100,000	This work
	L (Rhomb_A7,Bi,Fe:VA;0)	22,800	[5]
Ce_2Bi	L (Ce_2Bi,Ce:Bi;0)	$0.667*GHSER_{Ce}+0.333*GHSER_{Bi}-81819.61+10.69*T$	This work
Ce_5Bi_3	L (Ce_5Bi_3,Ce:Bi;0)	$0.625*GHSER_{Ce}+0.375*GHSER_{Bi}-91773.33+12.28*T$	This work
Ce_4Bi_3	L (Ce_4Bi_3,Ce:Bi;0)	$0.571*GHSER_{Ce}+0.429*GHSER_{Bi}-100319.44+12.71*T$	This work
CeBi	L (CeBi,Ce:Bi;0)	$0.5*GHSER_{Ce}+0.5*GHSER_{Bi}-110541.34+18.19*T$	This work
$CeBi_2$	L ($CeBi_2$,Ce:Bi;0)	$0.333*GHSER_{Ce}+0.667*GHSER_{Bi}-81804.87+14.18*T$	This work
Ce_2Fe_{17}	L (Ce_2Fe_{17},Ce:Fe;0)	$0.125*GHSER_{Ce}+0.895*GHSER_{Fe}-9684.616+5.672*T$	[4]
$CeFe_2$	L ($CeFe_2$,Ce:Fe;0)	$0.33*GHSER_{Ce}+0.67*GHSER_{Fe}-8941.667+3.284*T$	[4]

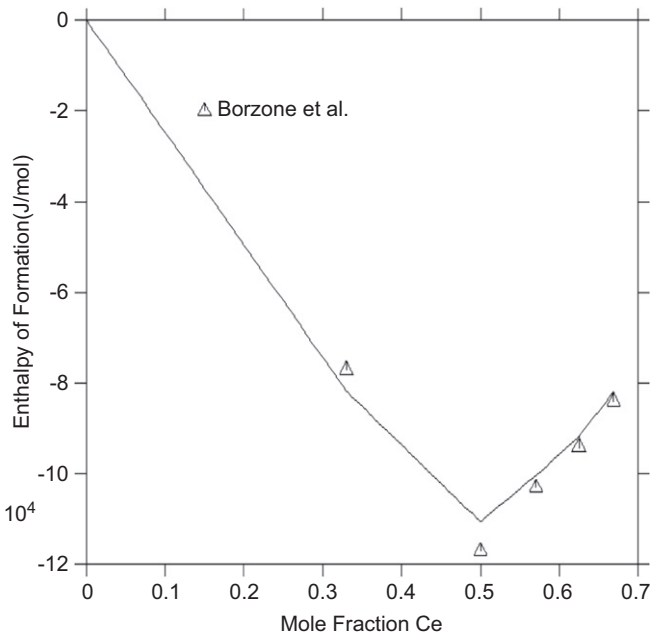

Fig. 4. Calculated enthalpy of formation of the Ce–Bi intermetallic compounds at 300 K in comparison with the experimental data [10].

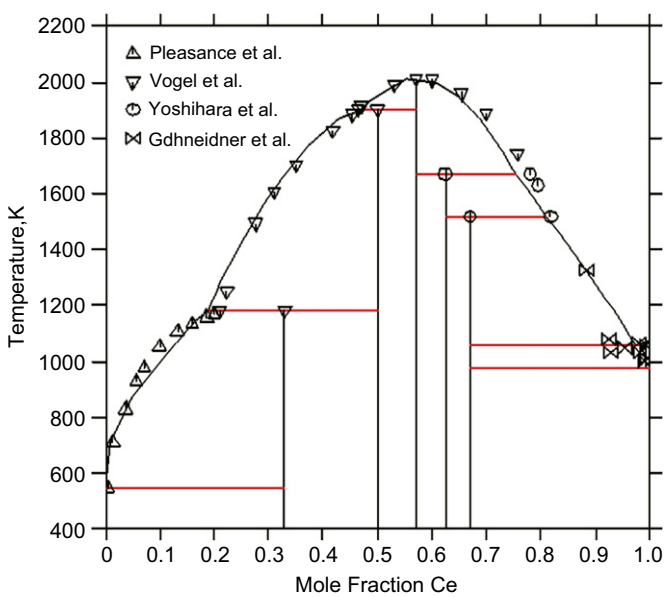

Fig. 5. Calculated Ce–Bi phase diagram in comparison with the experimental data [7–9].

4. Results and discussion

4.1. Thermodynamic modeling and calculated phase diagram of the Ce–Bi binary system

By using the module PARROT of the software Thermo-Calc, the thermodynamic parameters of the Ce–Bi binary system were optimized on the basis of the available information about phase equilibria and thermodynamics in literatures. The thermodynamic parameters obtained for the constituent binary systems are presented in Table 2.

The enthalpies of mixing of liquid Ce–Bi alloys were assessed as a first step in obtaining suitable parameters for liquid. As illustrated in Fig. 3, the enthalpy of mixing of the liquid alloys was calculated for comparison with the Miedema's model predictions of Dong et al. [11]. As can be seen, mixing enthalpy as assessed in this study indicates a general tendency similar to that estimated using Miedema's model.

The formation enthalpies of the intermetallic compounds measured by Borzone et al.[10] and the invariant reactions of the Ce–Bi binary system were adopted to optimize thermodynamic parameters of Ce_2Bi, Ce_5Bi_3, Ce_4Bi_3, CeBi and $CeBi_2$. The calculated enthalpy of formation of the intermetallic compounds at 300 K is shown in Fig. 4, compared with the reported data [10]. It is obvious that reasonable agreement was realized between the experimental data and the assessment.

Fig. 5 demonstrates the refined phase diagram of the Ce–Bi binary system in comparison with the available experimental data. In combination with Table 3, which lists all invariant reactions in the Ce–Bi system, it can be seen that the calculated phase equilibria are in good agreement with all experimental data [6].

4.2. Extrapolation of the Fe–Ce–Bi ternary system

No ternary phase in the Fe–Ce–Bi ternary system has yet been reported. By combining the Ce–Bi binary system in the present study with the Ce–Fe [4] and Bi–Fe [5] binary systems previously optimized, the Fe–Ce–Bi ternary system was extrapolated directly. Fig. 6 shows the calculated Fe–Ce–Bi isothermal section at 773 K. There are eight three-phase regions which are Fe–$CeBi_2$–liquid, Fe–CeBi–$CeBi_2$, Fe–Ce_4Bi_3–CeBi, Fe–Ce_2Fe_{17}–Ce_4Bi_3, Ce_5Bi_3–Ce_4Bi_3–Ce_2Fe_{17}, Ce_5Bi_3–Ce_2Fe_{17}–$CeFe_2$, $CeFe_2$–Ce_5Bi_3–Ce_2Bi and Ce–Ce_2Bi–$CeFe_2$ phase regions. Additionally, the liquidus projection was further calculated as shown in Fig. 7 and the calculated invariant reactions and temperatures involved liquid in the Fe–Ce–Bi ternary system are summarized in Table 4. Further experimental data are needed to verify the calculation.

4.3. Explanation of interfacial reaction in the Fe–Ce/Bi couple

4.3.1. Phase equilibria and reaction path

In order to check whether or not the extrapolated phase equilibria are reasonable, the diffusion path between the Fe–Ce alloy and the Bi metal is further investigated. Fe–Ce/Bi diffusion couples were fabricated by loading a piece of homogenized Fe-45 at%–Ce plate and a pure Bi bulk in silicon capsules back-filled with argon, then the capsules were held in a furnace at 773 K for 30–120 min. After that, the capsules were removed from the furnace and quenched in room temperature water. Because the holding temperature of 733 K is higher than the melting point of Bi (544 K) but lower than the eutectic temperature of Fe–Ce binary system (865 K), the Bi melts and surrounds the Fe-45 at%–Ce alloys plate to form a solid/liquid diffusion couple (Fig. 8).

The microstructure of the Fe–Ce/Bi couple was examined via electron probe microanalysis (EPMA) using a JEOL JXA-8800R (Japan Electron Optics Ltd., Tokyo, Japan) equipped with a wavelength-dispersive spectrometer. Two reaction layers seem to exist in these Fe-45 at%–Ce/Bi couples. A backscattered electron (BSE) image of the Fe–Ce/Bi couple for 70 min is shown in Fig. 7. According to the EPMA,

Table 3
Invariant reactions in binary Ce–Bi system: comparison between calculation and experimental.

Reaction	X_{Ce} of phase			Calculated temperature (*K*)	Literature temperature (*K*)	Reference
Liquid = $CeBi_2$ + Rhombo_A7	0.005	0.33	0.001	544.5	544	[8]
Liquid + CeBi = CeBi2	0.21	0.5	0.33	1183	1183	[7]
Liquid + Ce_4Bi_3 = CeBi	0.465	0.571	0.5	1901	1903	[7]
Liquid = Ce_4Bi_3	0.571	0.571		2013	2013	[7]
Liquid + Ce_4Bi_3 = Ce5Bi3	0.78	0.571	0.625	1668	1670	[9]
Liquid + Ce_5Bi_3 = Ce2Bi	0.818	0.625	0.67	1516	1518	[9]
Liquid = BCC_A2 + Ce_2Bi	0.93	0.982	0.67	1056	1032	[10]
BCC_A2 = FCC_A1 + Bi_2Ce	0.99	0.995	0.67	976.4	999	[10]

the reaction layer adjacent to the Fe–Ce alloy contains Fe+Bi+$CeBi_2$, and the next layer also includes $CeBi_2$+Bi+Fe. The difference lies in the morphology of these two layers, as shown in Fig. 7. It is easy to see that these two layers are virtually one diffusion layer. Considering that Bi exists as a liquid at 773 K, Bi should correspond to the liquid in the reaction. Furthermore, small quantities of needle-like $CeBi_2$ were embedded in the Bi end-member of the couple and these pieces of $CeBi_2$ should be products of the remaining Bi-enriched liquid. Therefore the diffusion path at 773 K should be Bi(liquid)/liquid+Fe+$CeBi_2$/Fe–Ce. According to van Loo et al. [16,17], it is impossible to develop three-phase areas in a ternary diffusion zone. The liquid+Fe+$CeBi_2$ three-phase layer in the couple might be resulted from liquid flowing into a $CeBi_2$–Fe two-phase layer. Nevertheless, equilibria of Fe+$CeBi_2$, liquid+Fe, and liquid+Fe+$CeBi_2$ can be deduced. In addition, $CeBi_2$ contains 68.51 at% Bi and 31.42 at% Ce and almost no Fe is solved in this compound. Clearly, the present experiment confirms the previously calculated phase equilbira, at least in the Bi-rich region of the Fe–Ce–Bi ternary system.

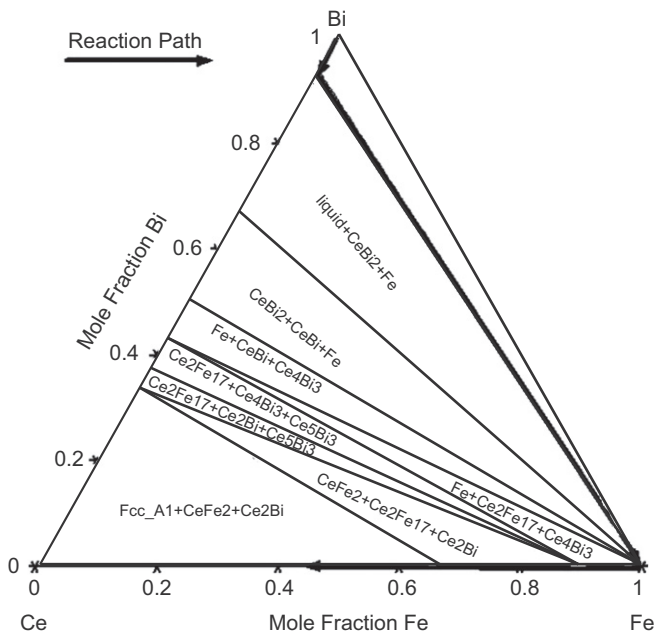

Fig. 6. Calculated Fe–Ce–Bi isothermal section at 773 K.

4.3.2. Formation of $CeBi_2$ in diffusion layer

It is interesting that only one intermetallic compound (IMC), $CeBi_2$, can be detected in Fe-45 at%–Ce/Bi couples no matter reacts for 30 min or for 120 min. Next, we will try to explain the formation of $CeBi_2$ using the assessed thermodynamic parameters. There is no intermetallic compound in the Fe–Bi binary system. Moreover, it is assumed that no ternary phase is formed in this ternary system, therefore, the Fe–Ce/Bi couple can be simplified to a Ce/Bi couple for the analysis of the reaction product at the interface.

Recently, Liu et al. [18] deduced a criterion to predict IMC formation during interfacial reaction. This criterion regards the interfacial reaction as a thermodynamic and kinetic process, which is naturally dependent on the driving force of nucleation and atomic diffusion across the interface. The criterion was given as

$$y_1 = -\frac{b}{64m^2}\left[\ln\frac{x_B^{\alpha}}{x_B^{X}x_B^{\beta}}\right]^3 - 3|\Delta x| \tag{12}$$

where b includes the effect of particle shape and contact angles on both sides of the IMC connecting to the parent phases and is assumed to be 20. The parameter m is the dimensionless driving force equal to $\Delta G_d/RT$, and Δx is the difference between the IMC and the average composition of the interface. Greater the y_1, the greater the volume of the IMC, indicating the compound will form ahead of other phases. This criterion wins its success in explaining the interfacial reactions, e.g. between Sn and Co [19]. By using the present thermodynamic parameters, parameters occurring in Eq. (12) were calculated and listed in Table 5. Then the criterion values of all possible compounds may be calculated, as listed in Table 6. It is readily apparent that $y_{1(CeBi2)}$ is the greatest, so $CeBi_2$ should form ahead of other compounds in the Fe-45 at%–Ce/Bi couple. Clearly, the prediction is in agreement with observation. This implies, at least partly, that the thermodynamic parameters assessed in this study for the Bi–Ce binary system are reasonable.

5. Conclusion

On the basis of the experimental data, the Ce–Bi binary system was thermodynamically optimized. Phase relations in the Fe–Ce–Bi

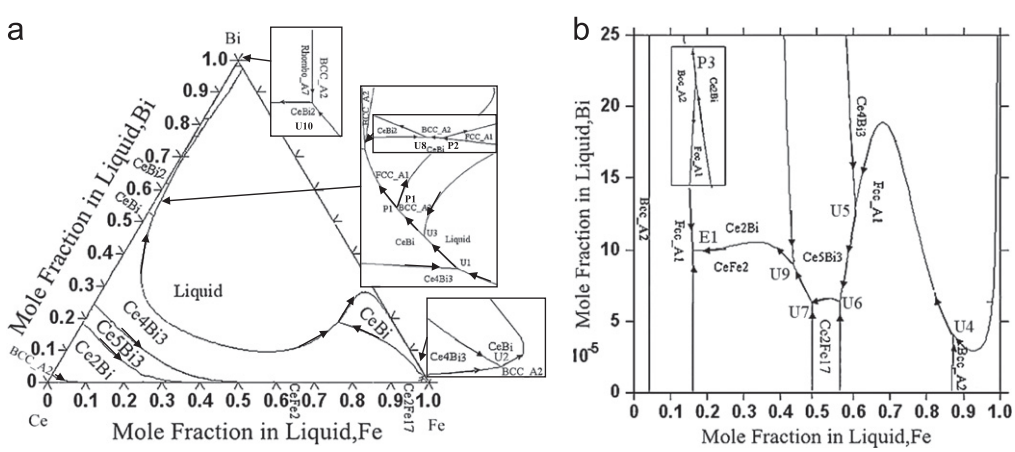

Fig. 7. (a) Calculated liquidus projection of the Fe–Ce–Bi ternary system (b) detailed view of the Fe–Ce side.

Table 4
Calculated invariant reactions and temperatures in the Fe–Ce–Bi ternary system.

Reaction	Type	T (K)	Reaction	Type	T (K)
Liquid(1#)+Ce_4Bi_3=Liquid(2#)+CeBi	U1	2191.69	Liquid+Ce_2Fe_{17}=Ce_5Bi_3+$CeFe_2$	U7	1213.91
Liquid(1#)+Ce_4Bi_3=CeBi+BCC_A2	U2	1891.83	Liquid+CeBi+FCC_A1=BCC_A2	P2	1183.49
Liquid(1#)+Liquid(2#)=BCC_A2+CeBi	U3	1809.35	Liquid+CeBi=BCC_A2+$CeBi_2$	U8	1183.09
Liquid+BCC_A2=Ce_4Bi_3+FCC_A1	U4	1672.96	Liquid+Ce_5Bi_3=Ce_2Bi+$CeFe_2$	U9	1176.75
Liquid+BCC_A2+CeBi=FCC_A1	P1	1667.28	Liquid+Ce_2Bi+FCC_A1=BCC_A2	P3	975.63
Liquid+Ce_4Bi_3=Ce_5Bi_3+FCC_A1	U5	1399.7	Liquid=Ce_2Bi+$CeFe_2$+FCC_A1	E1	867.98
Liquid+FCC_A1=Ce_2Fe_{17}+Ce_5Bi_3	U6	1300.55	Liquid+BCC_A2=$CeBi_2$+Rhombo_A7	U10	544.58

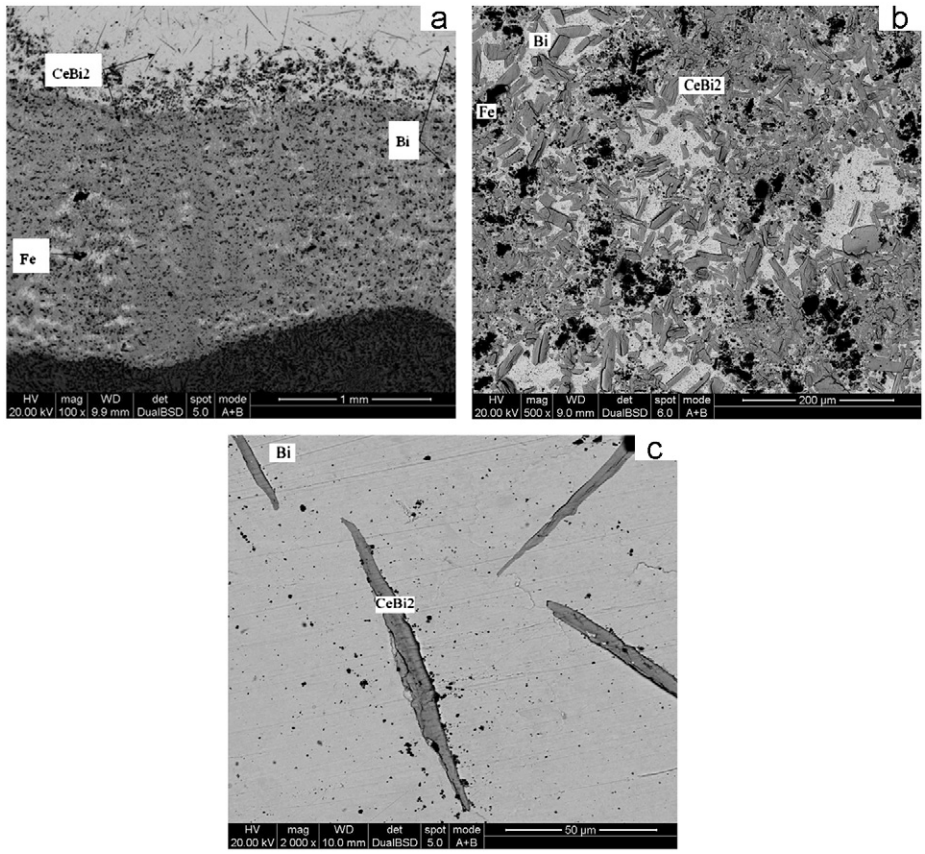

Fig. 8. BSE images of Fe-45 at%–Ce/Bi couples reacted at 773 K for (a) 70 min; (b) magnified area of the reaction layer; and (c) magnified area of the liquid phase.

Table 5
Calculated parameters in Liu et al. [18] model for diffusion couple Fe-45 at%–Ce/Bi at 773 K.

System	Temperature (K)	Phase	m (G_d/RT)	x_{Bi}^{FCC}	x_{Bi}^{liquid}	x_{Bi}^{x}	$\lvert\Delta x\rvert$
Fe-45 at%–Ce/Bi or FCC(Ce)/liqud(Bi)	773	Ce_2Bi	2.030	0.001	0.599	0.33	0.030
		Ce_5Bi_3	3.325	0.001	0.599	0.375	0.075
		Ce_4Bi_3	4.528	0.001	0.599	0.429	0.129
		CeBi	5.360	0.001	0.599	0.5	0.200
		$CeBi_2$	1.133	0.001	0.599	0.667	0.367

Table 6
Comparison of experiment and prediction by Liu et al. [18] models on IMC formation in the diffusion couple Fe-45 at%–Ce/Bi at 773 K. Thermodynamic parameters used in the calculation are from the present work.

System	Phase	Driving force (J/mol)	Criterion [24] y_1	IMC forming first	
				Predicted	Observed
Fe-45 at%–Ce/Bi or FCC (Ce)/liquid (Bi)	Ce_2Bi	12754.72	11.115	$CeBi_2$	$CeBi_2$
	Ce_5Bi_3	20893.84	4.261		
	Ce_4Bi_3	28454.61	2.216		
	CeBi	33682.17	1.416		
	$CeBi_2$	7123.28	51.168		

ternary system were reasonably extrapolated. The reaction path, liquid Bi/$CeBi_2$ + Fe/Fe-45 at% Ce, in Fe-45 at% Ce/Bi diffusion couples at 773 K was determined, and it was found that only one intermetallic compound, $CeBi_2$, was formed at the interface in this couple. The experimental results were reasonably explained.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2012.05.004.

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